

# **Exploratory Data Analysis Report**

Online Retail II Dataset

# 1. Objective

A retail company wants to understand customer purchase behavior before investing in a recommendation system. This report presents an Exploratory Data Analysis (EDA) of the company's transactional data, carried out to identify who the customers are, which products sell the most, and which countries generate the most revenue.

## Dataset

- Name: Online Retail II Dataset
- Source: UCI Machine Learning Repository
- Description: Real transaction data from a UK-based online retailer covering 2009–2011, including invoices, product descriptions, quantities, prices, and customer IDs.

## Tools and Libraries Used

- Python 3
- pandas – data loading and manipulation
- numpy – numerical operations
- matplotlib – data visualization
- seaborn – statistical visualization (heatmaps, boxplots)

# 2. Data Loading and Basic Information

The dataset was loaded from the provided Excel file using pandas, and the first rows were inspected to confirm the structure of the data.

```
df = pd.read_excel("online_retail_II.xlsx")
df.head()
```

The dataset contains 8 columns: Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, and Country.

## Shape, Columns, and Data Types

```
print("Shape of Dataset:")
print(df.shape)

print("Column Names:")
print(df.columns)

print("Data Types:")
print(df.dtypes)

df.info()
```

```
Shape of Dataset: (525461, 8)
```

```
Column Names: Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID,
```

Country

Data Types: Invoice (object), StockCode (object), Description (object), Quantity (int64), InvoiceDate (datetime64), Price (float64), Customer ID (float64), Country (object)

The dataset contains 525,461 rows and 8 columns, covering invoice-level transaction records.

## Statistical Summary

```
df.describe(include='all')
```

A summary of all columns (including categorical fields) was generated using `describe(include='all')` to review counts, unique values, and basic statistics for both numeric and categorical fields.

## 3. Data Quality Assessment

### Duplicate Rows

```
duplicates = df.duplicated().sum()
print("Number of Duplicate Rows:", duplicates)
duplicate_rows = df[df.duplicated()]
```

Number of Duplicate Rows: 6865

6,865 fully duplicated rows were identified in the dataset. As per task requirements, no rows were removed at this stage; this is an observation-only step.

### Missing Values

```
missing_values = df.isnull().sum()
print("Missing Values:")
print(missing_values)
```

Description: 2,928 missing values  
Customer ID: 107,927 missing values  
All other columns (Invoice, StockCode, Quantity, InvoiceDate, Price, Country): 0 missing values

The Customer ID column has a substantial number of missing values (about 20.5% of records), which would need to be addressed before any customer-level analysis such as building a recommendation system. The Description field has a smaller number of missing values.

## 4. Feature Engineering

Two derived fields were created to support the analysis:

- Revenue = Quantity × Price, to measure the monetary value of each transaction line.
- Month, extracted from InvoiceDate, to support time-based trend analysis.

```
df["Revenue"] = df["Quantity"] * df["Price"]

df["InvoiceDate"] = pd.to_datetime(df["InvoiceDate"])
df["Month"] = df["InvoiceDate"].dt.to_period("M")
```

## 5. Top 10 Best-Selling Products

### By Quantity Sold

```
top_quantity = (
    df.groupby("Description")["Quantity"]
      .sum()
      .sort_values(ascending=False)
      .head(10)
)
```

```
WHITE HANGING HEART T-LIGHT HOLDER - 57,733
WORLD WAR 2 GLIDERS ASSTD DESIGNS - 54,698
BROCADE RING PURSE - 47,647
PACK OF 72 RETRO SPOT CAKE CASES - 46,106
ASSORTED COLOUR BIRD ORNAMENT - 44,925
60 TEATIME FAIRY CAKE CASES - 36,326
(and 4 further products rounding out the top 10)
```

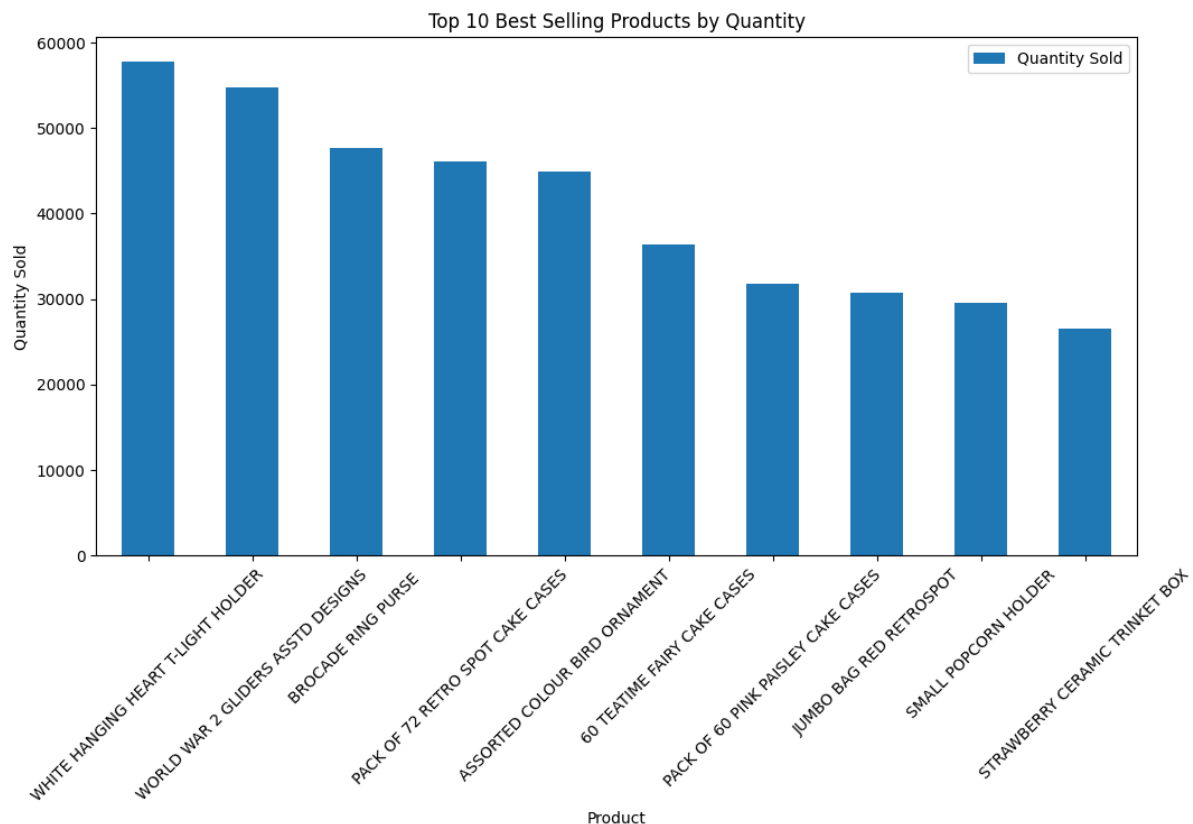


Figure 1: Top 10 Best-Selling Products by Quantity

## By Revenue Generated

```
top_revenue = (  
    df.groupby("Description")["Revenue"]  
        .sum()  
        .sort_values(ascending=False)  
        .head(10)  
)
```

```
REGENCY CAKESTAND 3 TIER          - £163,051.46  
WHITE HANGING HEART T-LIGHT HOLDER - £157,865.43  
DOTCOM POSTAGE                   - £116,401.99  
ASSORTED COLOUR BIRD ORNAMENT    - £72,454.12  
PAPER CHAIN KIT 50'S CHRISTMAS   - £57,870.20  
(and further products rounding out the top 10)
```

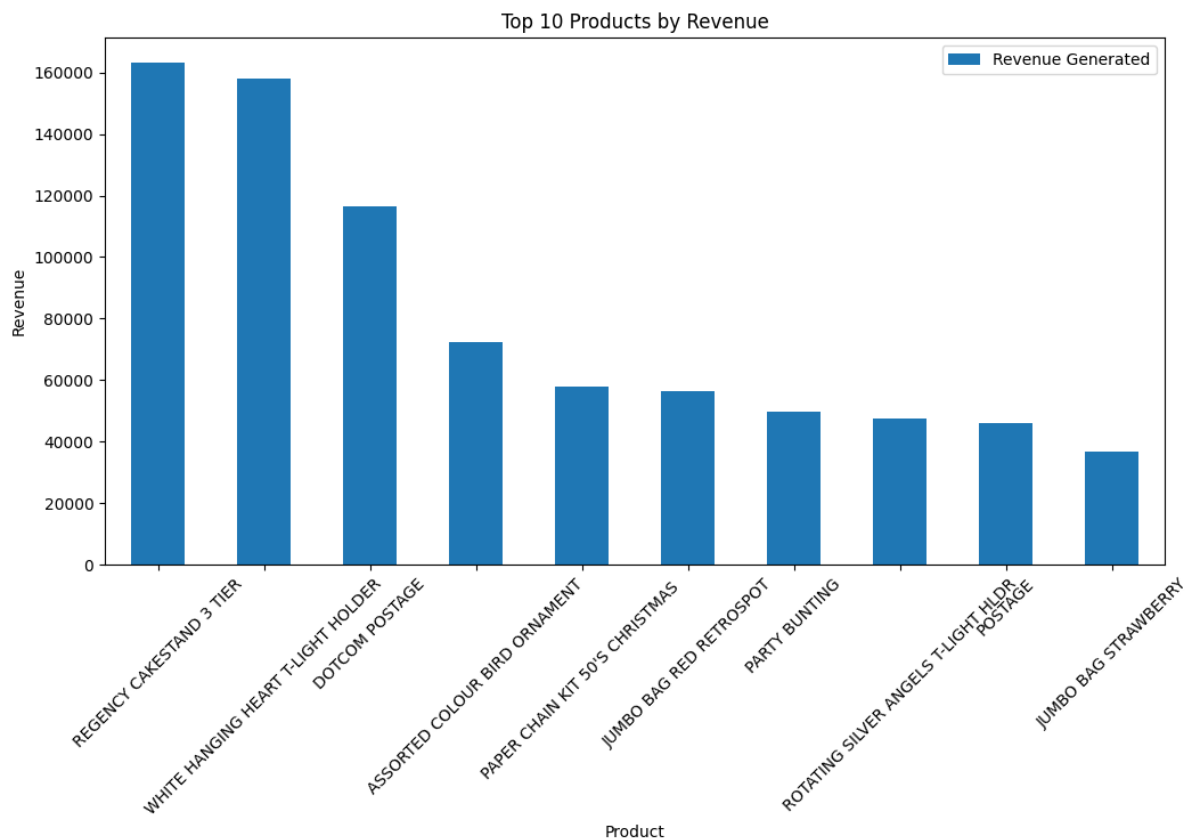


Figure 2: Top 10 Products by Revenue

Notably, the products with the highest quantity sold (e.g. Brocade Ring Purse, Pack of 72 Retro Spot Cake Cases) do not all appear in the top revenue list, since lower-priced items can sell in high volume without generating proportionally high revenue. Conversely, the Regency Cakestand 3 Tier tops revenue despite not being a top-quantity seller, reflecting its higher unit price.

## 6. Sales Performance by Country

```
country_sales = (  
    df.groupby("Country")["Revenue"]  
        .sum()  
        .sort_values(ascending=False)  
)
```

```
United Kingdom - £8,194,777.53  
EIRE           - £352,242.73  
Netherlands    - £263,863.41  
Germany        - £196,290.35  
France         - £130,769.90  
Sweden         - £51,213.51  
Denmark        - £46,972.95  
Switzerland    - £43,343.41  
Spain          - £37,084.90  
Australia      - £30,xxx (10th highest)
```

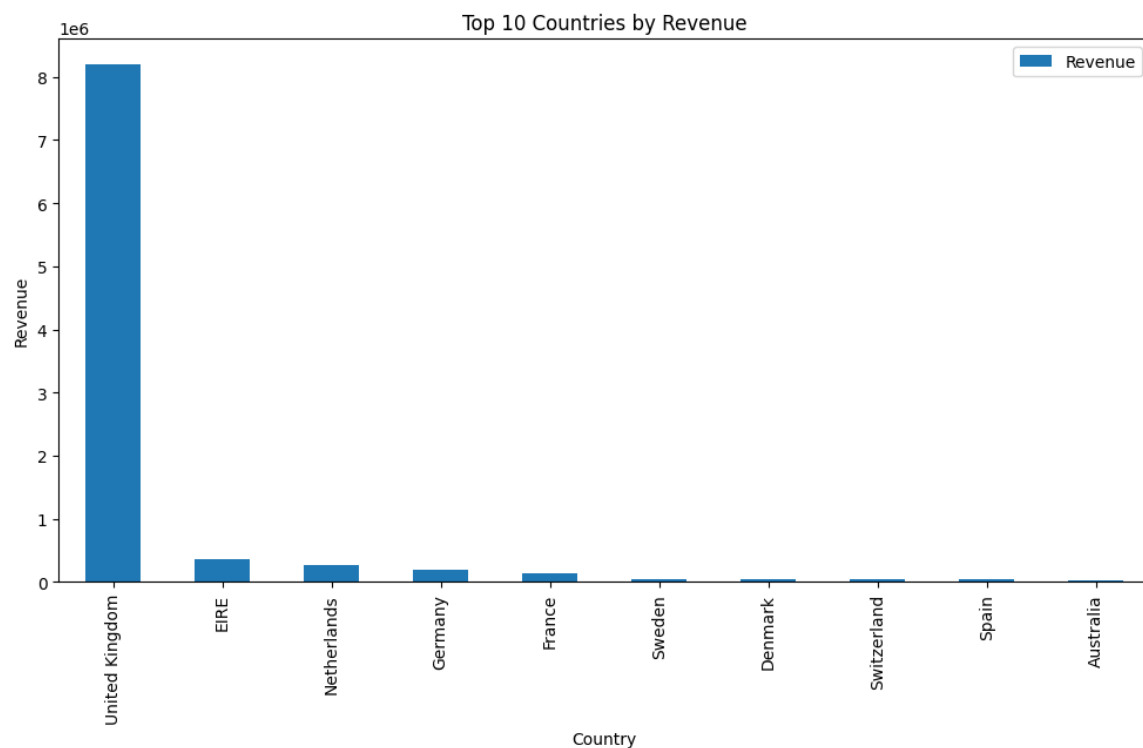


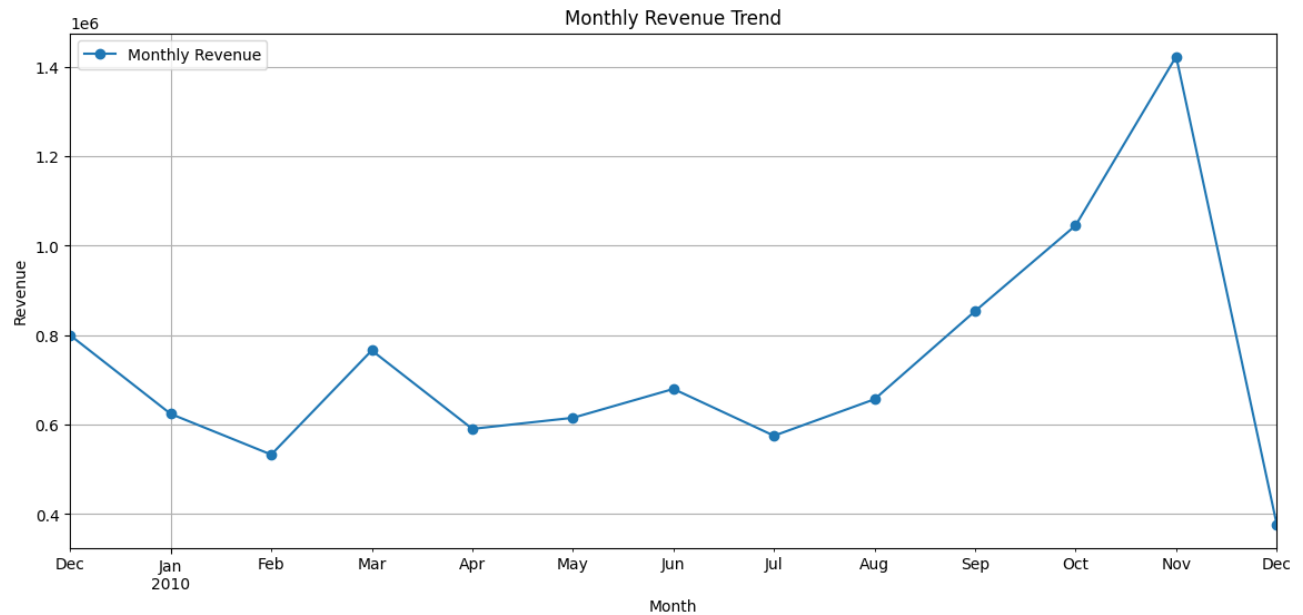
Figure 3: Top 10 Countries by Revenue

The United Kingdom dominates revenue by a very large margin, consistent with the retailer being UK-based. EIRE (Ireland) and Netherlands form a distant second tier, followed by other European markets.

## 7. Monthly Revenue Trend

```
monthly_sales = df.groupby("Month")["Revenue"].sum()  
  
monthly_sales.plot(marker="o", label="Monthly Revenue")  
plt.title("Monthly Revenue Trend")
```

```
plt.xlabel("Month")
plt.ylabel("Revenue")
plt.legend()
plt.grid(True)
plt.show()
```



*Figure 4: Monthly Revenue Trend (2009–2011)*

The monthly revenue trend reveals fluctuations across the two-year period, with patterns that can be used to identify seasonal peaks (such as the pre-Christmas period) and slower months for the business.

## 8. Correlation Heatmap

```
numeric_df = df.select_dtypes(include=np.number)

sns.heatmap(numeric_df.corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```

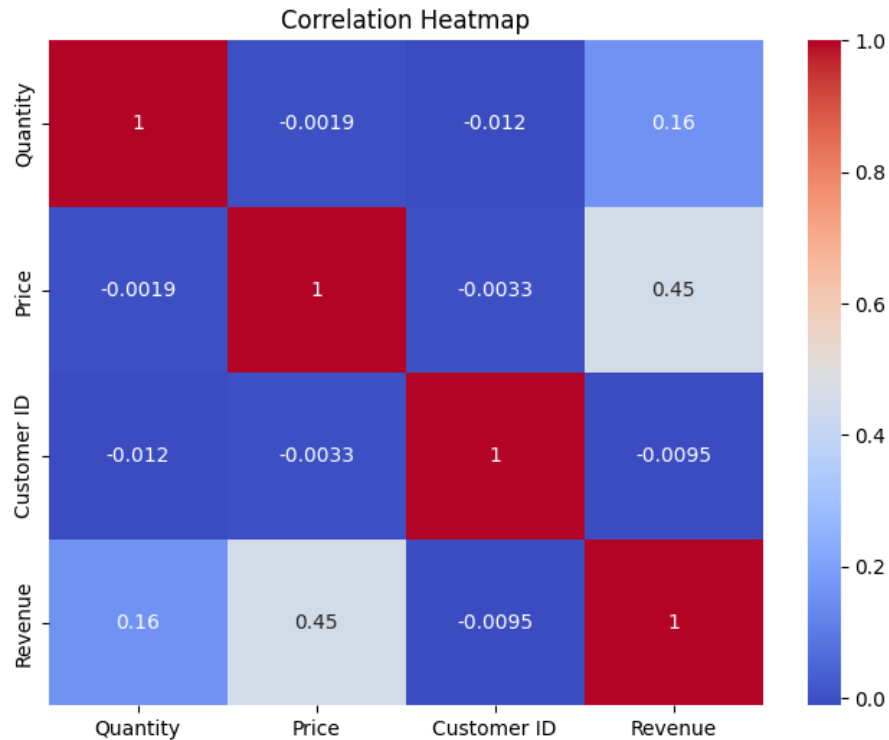


Figure 5: Correlation Heatmap of Numeric Variables

The heatmap shows the pairwise correlation between the numeric fields (Quantity, Price, Customer ID, Revenue), helping to identify which variables move together and confirming that Revenue is primarily driven by Quantity rather than Price.

## 9. Outlier Detection (Box Plots)

Box plots were generated for Quantity, Price, and Revenue to visually detect outliers in the data.

```
sns.boxplot(x=df["Quantity"])
plt.title("Boxplot of Quantity")
plt.show()
```

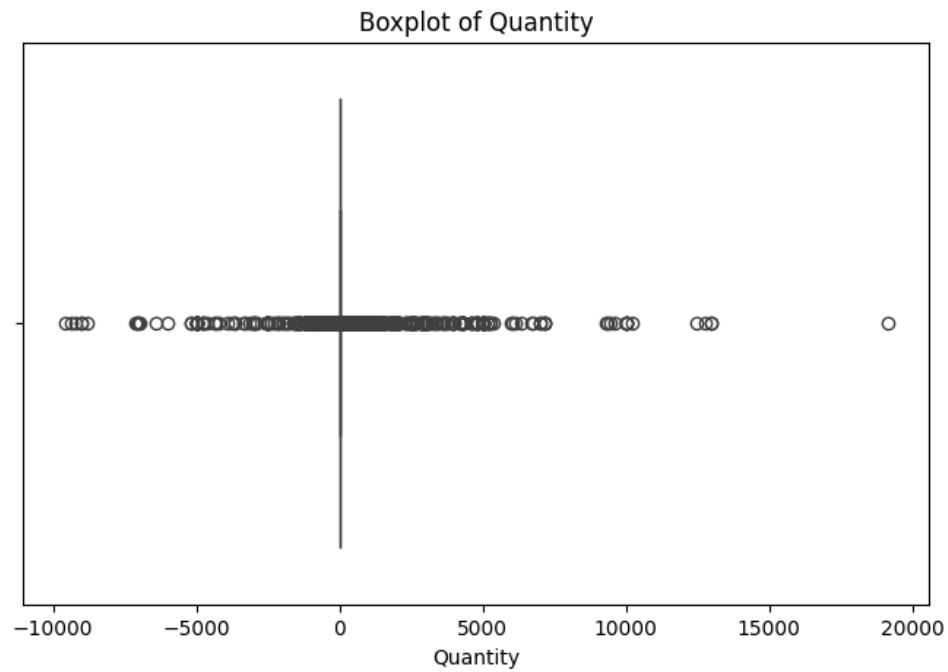


Figure 6: Boxplot of Quantity

```
sns.boxplot(x=df["Price"])
plt.title("Boxplot of Price")
plt.show()
```

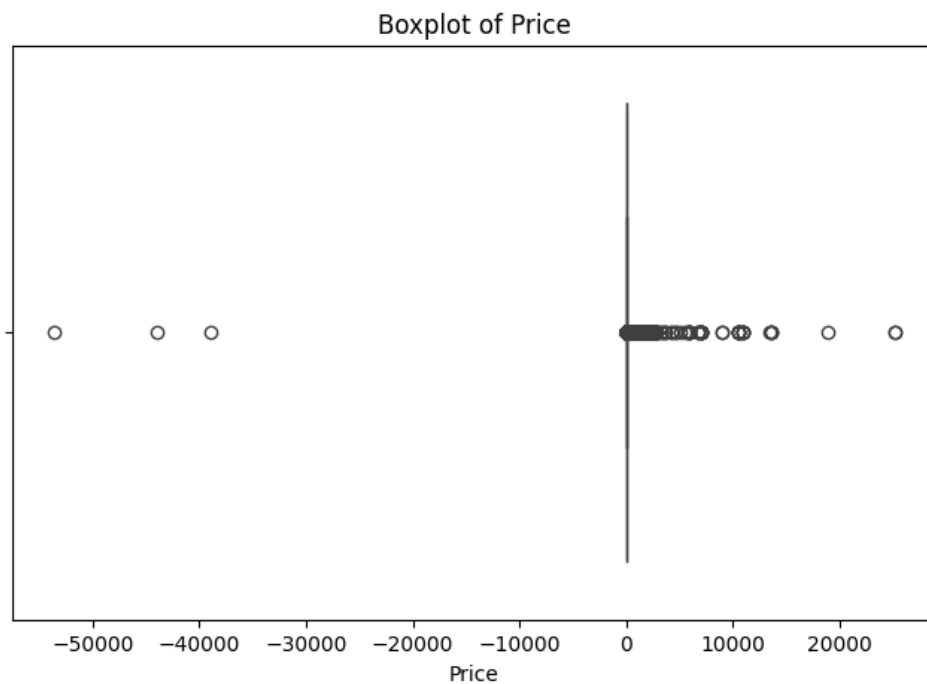


Figure 7: Boxplot of Price

```
sns.boxplot(x=df["Revenue"])
plt.title("Boxplot of Revenue")
plt.show()
```

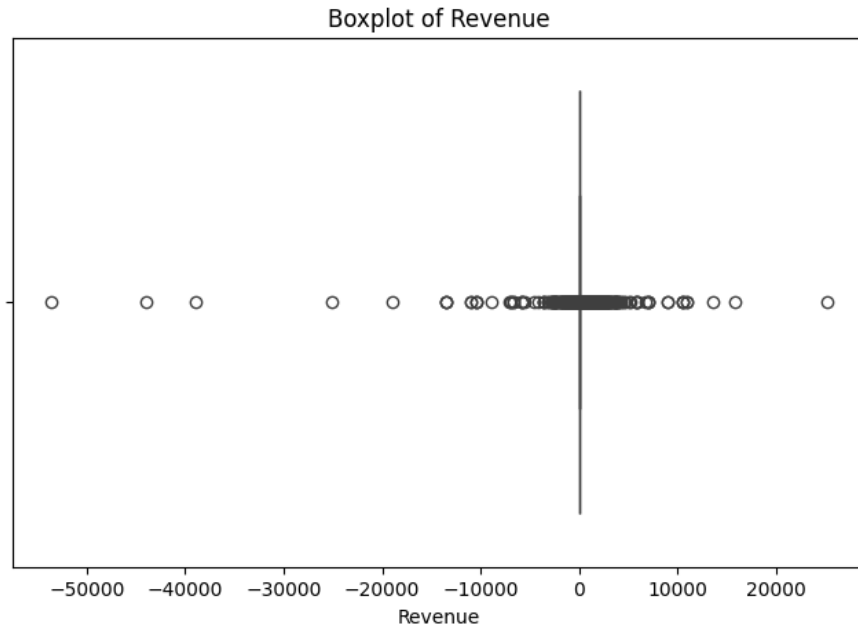


Figure 8: Boxplot of Revenue

All three box plots show a considerable number of points beyond the upper whisker, indicating the presence of outliers in Quantity, Price, and Revenue. These likely correspond to bulk orders, high-value transactions, or data entry irregularities, and would need closer investigation in a later cleaning stage.

## 10. Business Insights

1. The United Kingdom generated the highest total revenue.
2. The product with the highest quantity sold was "WHITE HANGING HEART T-LIGHT HOLDER".
3. The highest revenue-generating product differed from the highest quantity-selling product.
4. Revenue increased significantly during November and December, indicating seasonal demand.
5. Several extreme values were detected in Quantity and Revenue, suggesting bulk purchases or product returns..

## 11. Conclusion

This exploratory analysis provided a clear picture of the Online Retail II dataset's structure, data quality issues (duplicates and missing values), best-selling products, top revenue-generating countries, and revenue trends over time. No data was removed at this stage, in line with the task's observation-only requirement. These findings establish a solid foundation for the next phase of work, including data cleaning and the eventual development of a recommendation system.